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B.Tech. Degree IV Semester Special Supplementary Examination
February 2020

ME 15- 1406 MANUFACTURING PROCESSES

(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
 (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What are the methods which are used to compensate liquid shrinkage and solid shrinkage in sand casting?
- (b) What is the advantage of sand casting over die casting?
- (c) How can we test the moisture content of sand casting sand?
- (d) Which casting method is suitable for small sized components with dimensional tolerance under 20 microns? Justify the answer.
- (e) Differentiate between hot chamber and cold chamber die casting processes.
- (f) Differentiate between blanking and punching operations for sheet metal.
- (g) Explain the advantages of forging over other metal deformation processes.
- (h) Explain the principle of arc welding.
- (i) Differentiate between spot welding and seam welding.
- (j) Explain the process of Thermit welding?

PART B

(4 × 10 = 40)

- II. (a) With figure, explain the nomenclature of a sand casting mold. (7)
- (b) What is the significance of color coding in sand casting mold? (3)
- OR**
- III. (a) Explain the function of Chaplets and core prints in the sand casting mold. (5)
- (b) What do you mean by aspiration ratio of the Sprue? Derive it by applying Continuity equation. (5)
- IV. (a) Explain different product design considerations for reducing cost of a sand casted component. (5)
- (b) Explain different types of Riser systems in sand casting. (5)
- OR**
- V. (a) Explain with process of vacuum molding. (6)
- (b) What are the conditions to be satisfied while locating riser in a sand casting mold? (4)
- VI. (a) What is the significance of Recrystallization temperature of metals? (3)
- (b) Explain with figure the different kinds of roll arrangements in rolling operation. (7)
- OR**
- VII. (a) Differentiate between coining and embossing. (4)
- (b) Explain different types of forging operations with figure. (6)
- VIII. (a) Explain the process of Electron beam welding with figure. (4)
- (b) What are the advantages of reverse polarity arc welding over straight polarity arc welding? (3)
- (c) What is the function of flux coating on welding electrodes? (3)
- OR**
- IX. (a) What are the different welding defects? State the methods for avoiding each of them. (6)
- (b) What are the advantages and disadvantages of Solid state welding processes when compared with Arc welding? (4)

